

Introduction to High Performance Computing and Optimization

4. part PVL

To get points for the parts of the PVL, the C or C++ code (or Python in MPI related tasks), the job scripts and, depending on the task, the times achieved must be submitted. The submission takes place via the upload in OPAL in the corresponding course element. Copying or typing code from fellow students is not permitted. You must write and submit your own code. The specified submission deadline must be adhered to. Your code will be tested. Faulty or plagiarized code and code that delivers significantly different times than specified will result in zero points for the PVL part. So make sure that the code has been compiled on the cluster and executed on the compute nodes. Code must be submitted in the file format intended for direct use (submissions as, e.g., docx or pdf will be discarded). Your code must run on by machine, no virtual environments. Everything you need to complete the tasks is already installed on the cluster.

Exercise 1

Consider the following C code snippet that processes sensor data from two arrays:

```
1  for (int i = 0; i < N; i++) {  
2      if (i % 2 == 0) {  
3          double v = sqrt(pow(a[i], 2) + pow(b[i], 2));  
4          processed[i] = v > 10.0 ? v : 0.0;  
5      } else {  
6          processed[i] = 0.0;  
7      }  
8  }
```

- (a) Optimize this code for better performance. Which optimization techniques do you apply?
- (b) Name two other techniques for serial code optimization.